RESEARCH IN HIGHER EDUCATION, Vol. 1
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PERSONALITY DEVELOPMENT AND THE COLLEGE EXPERIENCE*

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Studies of freshman-senior data on undergraduates show that, despite major differences among institutions and students, the direction of net change was basically the same in several diverse colleges. Institutional freshmen means spanned a wide range, yet students became more autonomous, more aware, more integrated, more aesthetically sensitive, more tolerant, more liberal in religious views, and less concerned about material possessions. Although all students tended to change in the same direction at all colleges, they did not become more similar; diversity increased for the total group and, more often than not, diversity increased within each college.

When sub-groups of similar students were examined within institutions, different patterns of change appeared, and were systematically related to such factors as college climate, student characteristics, teaching practices and study activities, and student-faculty relationships. Thus, differential change occurred as a function of institutional "fit." When there is a close "fit" change proceeds along lines indicated by cultural and genetic forces and is consistent among diverse colleges. Where there is a "misfit" – students who have not developed to the general level at which a college operates, or whose development has gone beyond the operating level of a college – significant changes occur which are strongly associated with varied college characteristics and educational practices.

In September, 1965, entering freshmen at 13 colleges with enrollments of 1500 or less completed the Omnibus Personality Inventory (OPI) during the orientation period at the beginning of the year. During spring, in 1969, those students still enrolled in 12 of the colleges, who were ready to graduate, completed the same instrument again. Some of these students also had been retested at the end of their freshman or sophomore years.

Two groups of studies have asked these data several questions concerning personality development during the college years and the effects colleges may, or may not, have on such development. Part One of this report describes the results from three sets of studies concerning "net change" – from analyses pooling students from all the colleges and from analyses pooling all students retested within each college. Studies comparing mean scores at entrance with mean scores at

*The research reported here was supported by NIMH Grant #MH14780-05

graduation asked, "Does change occur? If so, in what areas? At what colleges?" Similar comparisons of responses to individual items pursued these questions further, asking, "Do particular changes in attitudes, beliefs, and behaviors underlie the mean scores?" Studies comparing changes in the standard deviations asked, "Do students become more similar during the college years, or are they more different from one another as seniors than as freshmen?"

Part Two describes results from studies of change among selected subgroups of students, and studies of relationships between changes among students and institutional characteristics. Studies examining change for subgroups of students in different colleges who had similar scores at entrance asked, "Do similar students at different colleges change differently? Does the impact of college vary depending upon the characteristics a student brings to it?" Studies examining relationships between changes for subgroups of students and varied measures of institutional characteristics asked, "If changes occur, are they systematically related to differences in the characteristics of the colleges?"

PART ONE

Results from the studies of change must be considered in the light of three major sets of findings concerning institutional characteristics and the characteristics of students as entering freshman (Chickering, 1966, 1969):

> There are dramatic differences among the institutions: in goals, in general climate, in rules and regulations, in curriculum, teaching, and evaluation, in sense of community and student faculty relationships, and in the concrete experiences and behaviors of student and faculty.
>
> There are wide differences in the characteristics of the students entering these diverse colleges; institutional means on the OPI span more than two standard deviations on several scales, and questionnaire data from other instruments reveal similarly striking diversity.
>
> There is a close fit between the characteristics of the students and the characteristics of the colleges. Self-selection and admissions practices have operated with sufficient power that certain kinds of students enter certain Project colleges and not others, and little overlap occurs among colleges substantially different from one another.

What developmental outcomes follow from these associations between particular kinds of students and particular kinds of colleges?

Does Change Occur? In What Areas? At What Colleges?

Change occurs, in all but one area measured by the OPI, at all colleges. Further, the direction of change is basically the same for all colleges—despite the great diversity in the mean scores of entering groups, and despite the major differences in the colleges. It is similar to changes observed during the freshman

TABLE I. Mean OPI Scores for Students Pooled from Twelve Colleges 1965–1969

	All Students N=585		All Men N=254		All Women N=269	
Scale	F/65	S/69	F/65	S/69	F/65	S/69
Autonomy	48	55	48	55	47	55
Practical Outlook	52	47	52	47	51	47
Impulse Expression	48	52	49	54	45	50
Complexity	48	50	48	51	47	49
Estheticism	49	52	46	50	52	55
Thinking Introversion	48	50	47	50	49	50
Masculinity-Femininity	49	48	55	53	43	42
Personal Integration	50	54	51	54	50	54
Anxiety Level	49	50	50	52	48	49
Religious Orientation	46	50	47	51	45	49
Theoretical Orientation	46	47	48	49	43	44
Altruism	51	52	49	50	54	54
Social Extroversion	47	47	46	46	48	48
Response Bias	47	48	48	49	46	47

Note: Differences in underlined scores significant at the 0.05 level or beyond.

and sophomore years (Chickering et al. 1969).

The evidence, consistent among varied levels of analysis, is unequivocal. Table I reports the results when individual students are pooled from all the colleges and when men and women are treated separately. Dependent t tests reflected statistically significant change on 13 of the 14 scales; 2 were between the 0.05 and 0.01 levels, and 11 were beyond 0.01. Only the Social Extroversion scale showed no change. Significant change occurs for men on 12 scales and for women on 9. Men and women change in the same direction on every scale.

The regularity of change persists in the results for single colleges. Men and women continue to change similarly regardless of the college and regardless of their mean scores at entrance. Three of the colleges provided data for only one sex, but separate analyses for men and women were possible at 9 others, and there was no instance of statistically significant change in different directions; indeed, when nonsignificant changes are included, only 29 cases of a possible 126 (9 colleges times 14 scales) changed in atypical fashion. This similarity of changes among men and women simplifies reporting and interpretation because attention can be restricted to findings from analyses which combined both groups.

Table II documents the consistent directions of change among the colleges. One hundred fifty four comparisons yielded 17 mean differences significant between the 0.05 and 0.01 levels, and 68 beyond 0.01; of these 85 cases, only 2–

TABLE II. Mean OPI Scores for Individual Colleges, Same Students Tested in 1965–1969

College	N	Autonomy		Practical Outlook		Impulse Expression		Complexity		Estheticism		Thinking Introversion		Masculinity Femininity		Personal Integration		Anxiety Level		Religious Orientation		Theoretical Orientation		Altruism		Social Extroversion		Response Bias	
		65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69
WJB	33	42	46	55	51	42	45	43	46	44	49	44	49	50	50	54	56	52	53	37	37	43	45	51	53	47	49	47	50
Savior	38	42	49	57	52	42	49	44	45	47	49	45	55	51	49	49	52	49	49	39	41	45	41	50	49	43	46	48	47
Stonewall	49	42	47	56	54	53	55	47	48	45	48	43	47	53	52	45	53	46	50	48	49	45	47	43	48	46	49	43	48
Simon	76	44	53	54	58	42	47	43	49	48	53	48	51	48	47	54	57	51	52	38	44	42	45	55	54	49	48	48	50
Divinity	51	46	54	52	46	43	48	46	49	48	51	47	48	49	49	53	58	52	54	39	43	45	44	53	53	49	48	49	51
Friendly	61	43	50	54	51	44	46	45	45	47	51	45	46	47	46	50	55	49	51	42	45	42	43	52	52	48	48	46	47
Kildew	36	59	65	41	39	57	61	62	62	60	60	60	55	43	44	48	53	48	50	62	60	52	52	54	53	47	47	46	47
Classic	13	61	65	42	39	54	61	58	62	55	61	58	61	54	48	51	52	50	49	55	58	54	56	51	52	44	47	50	51
Elder	123	54	63	47	42	49	55	50	54	50	53	51	52	49	47	50	52	50	50	52	58	49	48	53	54	46	45	47	46
Woodbine	56	48	55	53	45	55	57	50	51	51	55	45	59	47	44	46	51	44	48	53	58	45	49	48	50	48	48	43	46
Rocket	20	44	51	54	52	50	55	47	50	40	43	41	45	60	57	51	50	49	48	49	53	51	51	43	46	44	47	51	51

Note: Differences in underlined scores significant at the 0.05 level or beyond.

in each case a different college and different scale– deviated from the typical direction. Ignoring statistical significance, and setting aside the Social Extroversion scale on which no general changes occurred, among 140 comparisons, only 16 were contrary to the usual direction. Seven of these contrary changes occurred at Kildew where mean scores at initial testing often were near the extremes, leaving little room for changes in the directions typical of most other students.

Students at all the colleges changed in the same direction on three scales, Autonomy, Practical Outlook, and Impulse Expression–and those changes also occurred consistently when men and women were examined separately. Eleven of the 12 colleges changed in the same direction on Personal Integration and Estheticism, and 10 reflected similar change on Complexity, Thinking Introversion, and Religious Orientation (Liberalism).

Figures 1 and 2, and the others appended, illustrate these findings for Autonomy and Estheticism and show the positions of the 12 colleges relative to each other and to the mean scores and average change when all colleges are combined. The results for Estheticism vividly reflect the consistency among the institutions. Even though they span a wide range of scores, and even though the amount of change is small, change at each college–except at Kildew where no change occurs–is close to the average.

Greatest changes occurred on Autonomy, Impulse Expression, Personal Integration, Estheticism, Complexity, Religious Liberalism, and Theoretical Orientation, where scores rose, and on Practical Outlook, where scores dropped. Smaller and less consistent changes occurred on Masculinity-Femininity, Altruism, Anxiety Level, and Response Bias. No consistent changes occurred on the Social Extroversion scale.

The consistencies across scales and colleges are somewhat exaggerated by the fact that some of the scales have items in common and some of them are intercorrelated.* So these measures are not entirely discrete and independent. But the general evidence supports three major conclusions: 1. Greater changes occur in some areas than in others, as described above. 2. Changes occur for both men and women on all the OPI scales. 3. These changes are highly consistent at all the colleges.

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*Systematic studies of OPI scale validity have been limited almost entirely to correlational analyses of relationships with other personality inventory scales. In addition this instrument shares the limitations of all personality inventories in being vulnerable to changes in verbal behavior or sophistication which may be unaccompanied by more generalized behavioral change. The consistent results from diverse colleges and diverse students, however, lends substantial support to the validity of these measures. Tables III, IV, V, and VI indicate the items which most contributed to scale score changes.

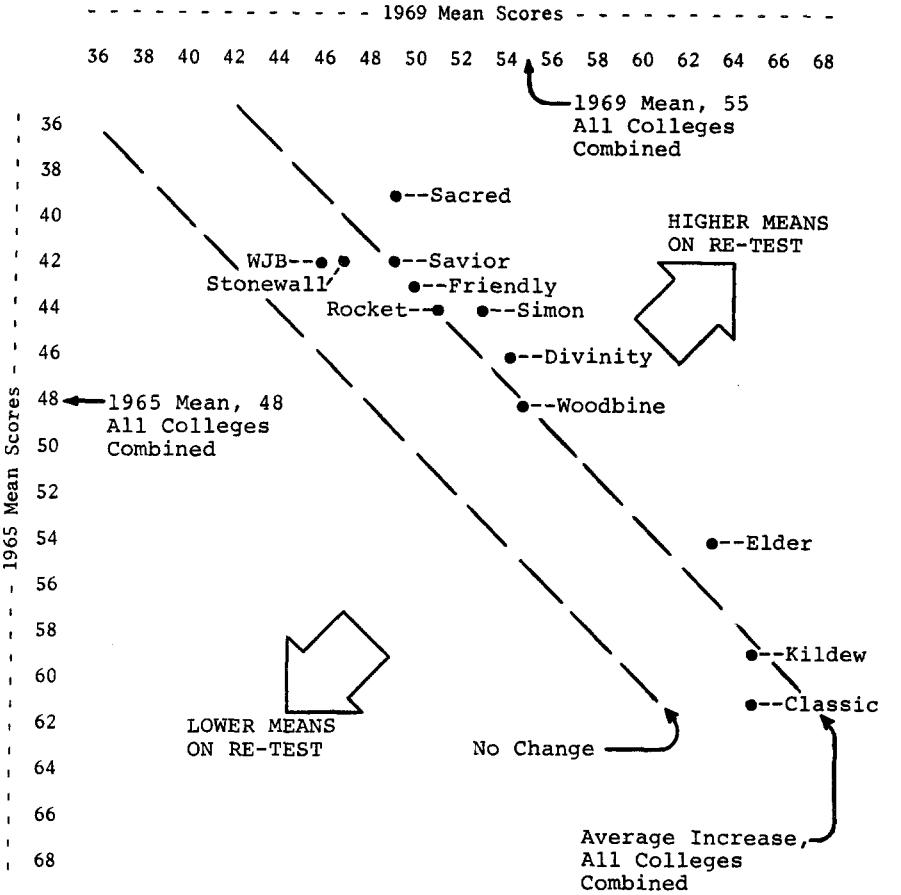

Fig. 1. AUTONOMY; 1965/1969 mean scores for individual colleges.

Do similarly consistent changes in particular attitudes, beliefs, and behaviors underlie the mean scores?

Item analyses were undertaken to see whether responses to individual items also changed consistently across the diverse colleges and students. These analyses indicated the proportions of students at each college whose responses as seniors had, or had not, changed. For each college, items on which 27% or more of the students had changed their responses were culled from the 390 which comprise the Inventory. For most colleges, this cut-off point produced between 15 and 25 "high change" items. This process identified particular attitudes, beliefs, or behaviors which changed for substantial numbers of students within each college.

Tables III, IV, and V give the items for a strongly conservative, church related

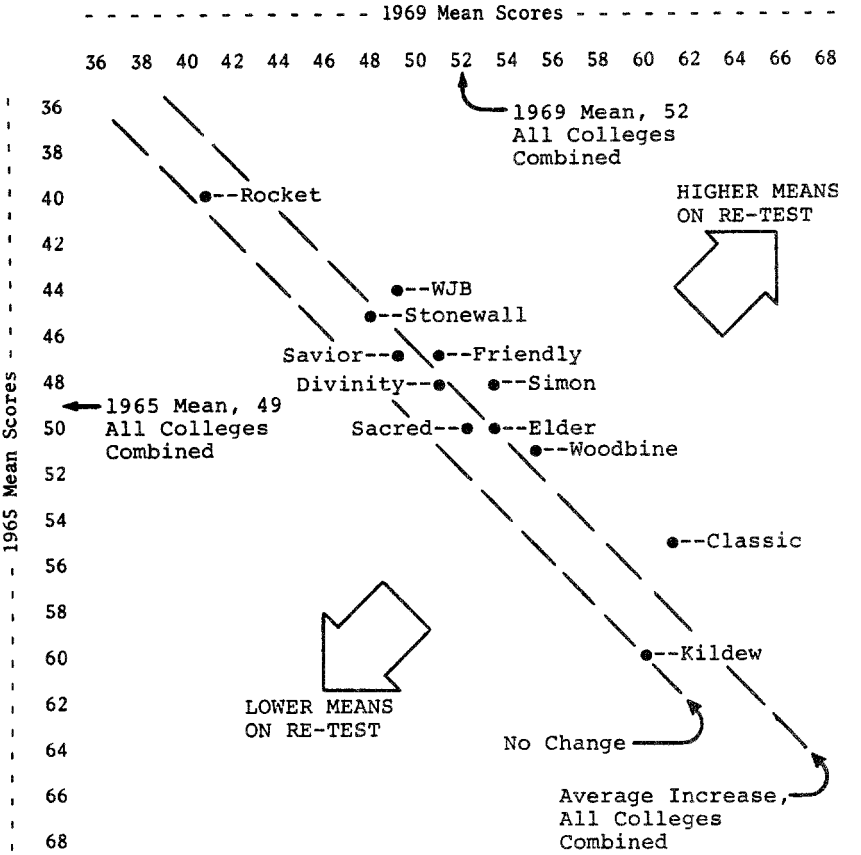

Fig. 2. ESTHETICISM; 1965/1969 mean scores for individual colleges.

college, a community college where most of the undergraduates are enrolled in engineering or business administration, and an elite liberal arts college with a liberal religious orientation. Many of the items are common for two of the colleges and some are common for three—and the directions of change are identical.

When these high change items were examined for all 12 colleges, 69 were common to 2 or more institutions; 30 items, listed in Table VI, were common to 3 or more. As would be expected, most of the items come from the scales on which greatest changes occurred. The significant point, however, is this: not only are the general changes consistent among the diverse students in the different colleges, as reflected by changes in scale score means, but also, changes in more particular attitudes, behaviors, and beliefs, are common to many of these diverse types of students and occur in many of the different types of colleges.

TABLE III. High Change Items for Simon, Conservative and Church-Related; N=76

Item	Percent responding "True" 65	69	Percent change
Perfect balance is the essence of all good composition.	72	29	47–
I believe it is a responsibility of intelligent leadership to maintain the established order of things.	80	38	42–
I like to talk about sex.	32	70	38+
Every person should have complete faith in a supernatural power whose decisions are obeyed without question.	83	46	37–
It is essential for learning or effective work that our teachers and leaders outline in detail what is to be done and how to do it.	66	30	36–
When I go to a strange city I visit art galleries.	12	46	34+
For most questions there is just one right answer, once a person is able to get all the facts.	50	17	33–
One needs to be wary of those persons who claim not to believe in God.	64	32	32–
One of my aims in life is to accomplish something that would make my mother proud of me.	67	36	31–
I prefer people who are never profane.	93	62	31–
Divorce is often justified.	16	47	31+
I disagree with statements and ideas expressed by my classmates or friends.	37	67	30+
Young people sometimes get rebellious ideas, but as they grow up they ought to get over them and settle down.	87	58	29–
One of the most important things children should learn is when to disobey authorities.	7	36	29+
When it comes to differences of opinion in religion we should be careful not to compromise with those whose beliefs are different from ours.	79	50	29–
I prefer social functions to which only a small group of intimate friends are invited.	55	84	29+
I like modern art.	29	58	29+
I would like to collect printing of paintings which I personally enjoy.	46	74	28+
I like worldliness in people.	13	41	28+
At times I have been so entertained by the cleverness of a crook that I have hoped he would get by with it.	26	53	27+
No man of character would ask his fiancee to have sexual intercourse with him before marriage.	91	64	27–
We should respect the work of our forefathers and not think that we know better than they did.	84	57	27–

TABLE IV. High Change Items for Rocket, Business and Engineering: N=20

Item	Percent responding "True" 65	69	Percent change
No man of character would ask his fiancee to have sexual intercourse with him before marriage.	80	30	50–
Young people sometimes get rebellious ideas, but as they grow up they ought to get over them and settle down.	95	45	50–
Nothing about communism is any good.	70	20	50–
The surest way to a peaceful world is to improve people's morals.	60	15	45–
Nothing about fascism is any good.	60	15	45–
I am active on the committees of school organizations.	10	50	40+
I tend to make decisions on the spur of the moment.	60	20	40–
I have never indulged in any unusual sex practices.	85	50	35–
Divorce is often justified.	45	80	35+
My church, faith, or denomination has the only true approach to God.	55	20	35–
I have never done any heavy drinking.	75	40	35–
I work under a great deal of tension.	15	45	35+
Only a fool would try to change our way of life in this country.	45	10	35–
I generally attend the meetings of school or community organizations.	30	60	30+
I do not like to appear on programs or to give oral reports.	75	45	30–
It is all right to get around the law if you don't actually break it.	35	65	30+
When I work on a committee I like to take charge of things.	30	60	30+
It is better never to expect much; then you are rarely disappointed.	60	30	30–
One of my aims in life is to accomplish something that would make my mother proud of me.	90	60	30–
I would enjoy writing a paper on the possible long-term effect or outcomes of a significant research discovery.	25	55	30+
I dislike test questions in which the information being tested is in a form different from that in which it was learned.	65	35	30–
It puzzles me why some people will so avidly read and discuss science fiction.	40	10	30–
Communism is the most hateful thing in the world today.	55	25	30–
Religion should be primarily a social force or institution.	20	50	30+

TABLE V. High Change Items for Elder, Elite Liberal Arts; N=125

Item	Percent responding "True" 65	69	Percent change
I believe it is a responsibility of intelligent leadership to maintain the established order of things.	66	19	47–
No man of character would ask his fiancee to have sexual intercourse with him before marriage.	53	8	45–
I go to church or temple almost every week.	70	16	44–
I have never done any heavy drinking.	86	46	40–
I generally attend the meetings of school or community organizations.	70	30	40–
I dislike women who disregard the usual social or moral conventions.	47	10	37–
I am active on the committees of school organizations.	53	17	36–
I should like to belong to several clubs or lodges.	54	18	36–
I would disapprove of anyone's drinking to the point of intoxication at a party.	70	34	36–
In the final analysis, parents generally turn out to be right about things.	81	46	35–
People would be happier if sex experience before marriage were taken for granted in both men and women.	25	60	35+
I like short, factual questions in an examination better than questions which require the organization and interpretation of a large body of material.	60	26	34–
I believe in a life hereafter.	66	33	33–
Perfect balance is the essence of all good composition.	43	10	33–
It is not the duty of a citizen to support his country right or wrong.	54	87	33+
I prefer people who are never profane.	46	13	33–
I believe women ought to have as much sexual freedom as men.	57	90	33+
Husbands, rather than wives, should have the final voice in family matters.	67	38	29–
There is nothing wrong with the idea of inter-marriage between different races.	66	94	28+
When science contradicts religion it is because of scientific hypotheses that have not been and cannot be tested.	37	9	28–
I do not like to see people carelessly dressed.	53	25	28–
I dislike test questions in which the information being tested is in a form different from that in which it was learned.	47	20	27–
The surest way to a peaceful world is to improve people's morals.	47	20	27–
My conversations with friends usually deal with such subjects as mutual acquaintances and social activities.	62	35	27–

TABLE VI. High Change Items Common to Several Colleges

Common Items	Direction of change*
Common to Eight Colleges	
No man of character would ask his fiancee to have sexual intercourse with him before marriage.	–
Common to Six Colleges	
I have never done any heavy drinking.	–
I believe it is a responsibility of intelligent leadership to maintain the established order of things.	–
Common to Five Colleges	
There is nothing wrong with the idea of intermarriage between different races.	+
I believe women ought to have as much sexual freedom as men.	+
Common to Four Colleges	
I prefer people who are never profane.	–
The surest way to a peaceful world is to improve people's morals.	–
Perfect balance is the essence of all good composition.	–
When science contradicts religion it is because of scientific hypotheses that have not been and cannot be tested.	–
In the final analysis, parents generally turn out to be right about things.	–
I like modern art.	+
I go to church or temple almost every week.	–
Common to Three Colleges	
Our way of doing things in this nation would be best for the world.	–
I like short, factual questions in an examination better than questions which require the organization and interpretation of a large body of material.	–
At times I have been so entertained by the cleverness of a crook that I have hoped he would get by with it.	+
I often feel that the people I meet are not interested in me.	–
Young people sometimes get rebellious ideas, but as they grow up they ought to get over them and settle down.	–
Every person should have complete faith in a supernatural power whose decisions are obeyed without question.	–
It is a pretty callous person who does not feel love and gratitude for his parents.	–
Every person ought to be a booster for his own home town.	–
Nothing about communism is any good.	–
I dislike test questions in which the information being tested is in a form different from that in which it was learned.	–
I dislike women who disregard the usual social or moral conventions.	–
Trends toward abstractionism and the distortion of reality have corrupted much art in recent years.	–
I like worldliness in people.	+
I never attend a sexy show if I can avoid it.	–
We should respect the work of our forefathers and not think that we know better than they did.	–
I like to talk about sex.	+
When I go to a strange city I visit art galleries.	+
I like to work late at night.	+

* + indicates more frequent agreement: – indicates less frequent agreement

Do students become more similar?

The standard deviations associated with initial and final testing show whether the distributions of individual scores on the various scales become narrower or wider after four years. Table VII indicates that, on every scale, standard deviations were larger after four years. Therefore, even though students at these colleges spanned a wide range as freshmen they had become even more diverse as seniors.

TABLE VII. Standard Deviations, 1965/1969; Students Pooled from Twelve Colleges; N=585

Scale	F/65	S/69	Difference
Autonomy	9.71	10.24	+0.53
Practical Outlook	8.60	9.34	+0.74
Impulse Expression	10.82	11.43	+0.61
Complexity	10.01	11.00	+0.99
Estheticism	9.55	10.13	+0.58
Thinking Introversion	10.05	10.38	+0.33
Masculinity-Femininity	9.70	9.90	+0.20
Personal Inegration	9.85	10.72	+0.87
Anxiety Level	9.76	10.16	+0.40
Religious Orientation	10.03	10.29	+0.26
Theoretical Orientation	9.37	9.87	+0.50
Altruism	9.51	9.87	+0.36
Social Extroversion	10.14	10.53	+0.39
Response Bias	9.13	9.64	+0.51

Changes in the standard deviations within each college are generally consistent with the changes when students were pooled. Table VIII reports 168 pairs of standard deviations—12 colleges times 14 scales. In 104 paris, standard deviations are higher after 4 years, 39 show no change, and 24 are lower. The 24 cases where lower standard deviations occur are scattered across all the scales, so increased similarity is not concentrated in a single area. Ten of the 24 cases occur at Kildew and Classic, the two colleges where students scored closest to the extremes as freshmen. "Ceiling effects" resulting from the nature of the instrument may account for the smaller standard deviations at these colleges. Individuals scoring close to the extremes at entrance could not move futher out but, because change typically is in their direction, more moderate classmates could move toward them, decreasing the differences reflected by the Inventory. Under these conditions of measurement it cannot be determined whether "true" changes toward similarity occured or whether the smaller standard deviations result from the

TABLE VIII. Standard Deviations for Individual Colleges, Same Students Tested in 1965–1969

College	N	Autonomy		Practical Outlook		Impulse Expression		Complexity		Estheticism		Thinking Introversion		Masculinity Femininity		Personal Integration		Anxiety Level		Religious Orientation		Theoretical Orientation		Altruism		Social Extroversion		Response Bias	
		65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69	65	69
WJB	33	8	8	7	9	8	8	8	9	9	10	9	10	9	11	9	8	9	9	3	5	7	8	9	7	11	10	8	8
Savior	38	6	8	7	9	9	11	9	9	9	9	10	11	10	10	11	12	10	12	6	7	9	10	9	10	11	11	9	10
Sacred	29	6	9	7	9	10	14	7	11	8	10	8	8	8	10	10	11	10	9	4	7	7	9	8	8	9	13	9	11
Stonewall	49	8	9	9	9	7	10	9	9	9	10	9	9	8	8	9	11	10	11	8	8	9	10	9	10	9	11	7	11
Simon	76	7	9	7	9	9	12	8	11	9	11	9	10	9	10	9	9	9	10	4	7	9	9	7	9	10	10	9	9
Divinity	51	7	7	7	7	10	11	8	11	9	10	9	10	10	9	8	9	9	8	5	7	9	10	10	12	9	10	8	9
Friendly	61	8	9	8	9	10	10	8	10	8	9	9	11	10	10	10	11	9	10	6	7	9	9	10	10	11	10	9	10
Kildew	36	11	9	6	7	11	9	11	8	8	8	8	9	7	8	8	12	9	10	6	6	10	9	9	11	10	9	6	7
Classic	13	7	5	8	7	8	9	10	12	8	9	10	7	11	11	9	11	10	13	10	7	11	9	10	10	9	11	8	8
Elder	123	8	7	8	7	10	10	9	10	9	9	9	10	10	10	10	10	11	11	9	7	9	9	9	9	10	10	10	10
Woodbine	56	8	10	7	8	11	12	9	12	8	10	10	11	8	10	8	11	9	10	8	8	8	11	9	11	9	10	9	10
Rocket	20	5	6	5	7	11	9	7	10	7	8	9	11	7	6	12	13	11	10	9	8	9	10	9	9	12	14	9	10

limitations of the instrument. Standard deviations did increase on seven scales at Kildew and on six at Classic, so in any event there was no highly generalized trend toward increasing similarity at these two colleges.

In general, students did not become more similar during their four years of college. Diversity increased for the total group and more often than not, diversity increased within each college.

Summary

Several different studies of net change generated these major findings:

1. Changes occur in most of the areas measured by the OPI.

2. The directions and amounts of change are similar for both men and women, and similar for the diverse groups of entering freshmen.

3. The directions and amounts of change also were similar at all the Project colleges, despite the dramatic differences among them.

4. The scales showing largest and most consistent changes were Autonomy, Impulse Expression, Personal Integration, Estheticism, Complexity, Religious Orientation (Liberalism), and Thinking Introversion, where scores rose, and Practical Outlook, where scores dropped. Smaller and less consistent changes occurred on Altruism, Theoretical Orientation, Anxiety Level, and Response Bias, where scores rose, and on Masculinity-Femininity, where scores dropped. No consistent changes occurred on Social Extroversion.

5. Changes in particular attitudes, beliefs, and behaviors also were common to many of the different types of students at the different colleges.

These findings suggest that there are several major vectors of personality development along which change occurs during the college years. Young adults probably pursue such development in whatever colleges they enter. In these small distinctive colleges, the fit between institutional characteristics and student characteristics at entrance permits development to proceed to about the same degree in each setting. For most of the students who attend, the differences among the colleges do not foster major differences in directions of development, nor in the areas where most development occurs. If distinctive influences occur, they must operate for only a minority of the students.

PART TWO

The several studies reported below primarily aimed to understand whether personality change in college is influenced by differences in goals, general climates, educational practices, and the concrete experiences and behaviros of students and faculty. At first the task seemed relatively simple–identify the important differences among the colleges, study the students when they enter and when they leave, and then look to see where the greatest changes occurred. But it turned out to be more complicated than that.

The first problem arose when we found that the different institutions attracted and enrolled very different kinds of students. At Savior and Simon there were no Kildew and Classic types, at Kildew and Classic no "Saviors" could be found, and between these extremes the mixtures varied at each college (Chickering, 1968). Under these circumstances, if differences in the amount of change occurred, who could say whether it stemmed from the differences among the institutions or from the equally substantial differences among the students? Still, we eagerly awaited the results from retesting completed at the end of the first or second years. Perhaps by the time those data were available a way would have been found to make sense of whatever differences in change appeared. But when the retest results became apparent our first problem was solved—after a fashion. Change apparently was the same at all the colleges. There were no striking differences, despite the differences among the colleges and the students. There was no need to study college impact. All the diverse approaches were equally effective—or ineffective (Chickering et al., 1969; Part One of this report).

But it was hard to believe that such dramatic institutional differences would not affect at least some of the students. Perhaps the masks of mean scores disguised individual changes within some of the groups. Several OPI scales were selected for more detailed study; on some statistically significant change had occurred and on some mean scores at both testings were identical. Individual scores at first testing were subtracted from retest scores, and the resulting distributions of individual change scores were examined.

The distribution of individual changes underlying these mean differences revealed four major patterns:

1. On some scales where significant increases had occurred, scores for practically all the students rose, but usually only a small amount.
2. On other scales where significant increases occurred, scores for a substantial number dropped, but relatively large increases among the majority outweighted the contrary minority.
3. On some scales where mean scores were identical at both testings, very few individuals had identical scores; many scores increased substantially, and many others dropped. Though the net effect was zero, many students had changed as much as half of a standard deviation or more.
4. Finally, of course, on several scales where means were identical, individual scores showed little change.

In effect, we had discovered three different patterns of college impact: (a) all students change in the same direction, (b) most students change in the same direction but a substantial minority changes in contrary fashion, and (c) most students change, but increases and decreases cancel each other to yield similar mean scores. These are among the patterns described by Feldman and Newcomb (1969) in their discussion of varied types of college impact.

These exploratory studies suggested that, despite the appearance of similar mean differences, changes at one college might be quite different from those at another. If that turned out to be the case, more effective study of college influences could be undertaken, and the Skager et al. (1965) approach suited our conditions well so it was adopted.

Do similar students at different colleges change differently?

Two studies examined change for subgroups of students at different colleges who had similar OPI scores at entrance. Attention focused on the five scales which reflected greatest change during the first two years. The first task was to find students from several different Project colleges whose entering scores were similar. With the small numbers of students entering the colleges to begin with, finding relatively narrow score intervals which included several respondents from diverse institutions was not easy. After scanning the distributions of individual scores at the different colleges, institutions and intervals were selected which optimized both institutional diversity and numbers of students. The process was further complicated because it seemed useful to examine both two year change and four year change, and to hold the score intervals constant while using different students and a slightly different group of colleges.

Table IX gives the score intervals finally used, and lists the two groups of six colleges with the numbers of students studied in each of the score intervals selected. As the figures indicate, in many cases the numbers for a particular college and scale are very small. But the first objective was achieved. Students with similar scores at entrance had been identified, and when the individual scores were averaged, groups with similar mean scores at entrance had been established. In no case did college means differ by more than two points, even though, for example, institutional means for all entrants at these colleges ranged from 42 to 59 on Autonomy and Impulse Expression, and from 43 to 62 on Complexity.

Next, retest means were computed for each subgroup to see whether the amount and direction of change would once again be similar across the diverse colleges. As the $X_1 - X_2$ figures in Table IX indicate, the differences between initial and retest scores varied substantially from college to college, both in the direction and amount of change. Further, the colleges maintained roughly similar positions in relation to one another on the different scales and for both two and four year change. Greatest increases in Complexity, Autonomy, and Impulse Expression and greatest decreases in Practical Outlook occur at Kildew. Elder students and the two year change groups at Classic share this pattern but show somewhat smaller changes. Simon, in comparison, consistently shows smaller increases and sometimes changes in a different direction; Stonewall, Friendly, and Divinity are associated with this pattern. Four year change on Estheticism is the one exception to the general relationships. There, Kildew's re-

TABLE IX. Change for Similar Students at Different Colleges

College	Scales and Score Intervals																			
	Estheticism 52–57				Complexity 57–62				Autonomy 53–58				Impulse Expression 52–57				Practical Outlook 42–47			
Two Year Change for Selected Subgroups																				
	N	65	67	X_1-X_2	N	65	67	X_1-X_2	N	65	67	X_1-X_2	N	65	67	X_1-X_2	N	65	67	X_1-X_2
Kildew	5	55	59	4	5	60	68	8	5	56	66	10	8	55	58	3	5	44	38	−6
Classic	5	54	59	5	5	59	57	−2	6	55	63	8	6	54	57	3	5	46	45	−1
Elder	14	55	58	3	10	59	57	−2	14	56	60	4	15	55	56	1	17	45	44	−1
Stonewall	11	55	54	−1	11	59	52	−7	10	54	50	− 4	26	55	57	2	6	46	48	2
Friendly	12	55	57	2	7	60	62	2	5	55	55	0	12	54	54	0	11	45	46	1
Simon	25	55	56	1	7	59	55	−4	7	54	55	1	9	54	50	4	17	46	47	1
Four Year Change for Selected Subgroups: (Same Scales and Score Intervals, different students)																				
	N	65	59	X_1-X_2	N	65	67	X_1-X_2	N	65	67	X_1-X_2	N	65	67	N_1-X_2	N	65	67	N_1-X_2
Kildew	5	56	56	0	8	60	64	4	11	56	65	9	9	55	62	7	11	44	38	−6
Elder	35	54	60	6	25	58	58	0	32	55	62	7	21	54	58	4	29	44	42	−2
Woodbine	25	55	57	2	8	59	59	0	13	54	58	4	7	55	58	3	8	46	40	−6
Friendly	17	54	57	3	5	60	50	−10	6	55	56	1	8	54	56	2	10	45	40	−1
Divinity	12	54	56	2	7	58	58	0	7	54	56	2	4	56	54	−2	9	45	43	−2
Simon	20	55	60	5	4	60	56	−4	9	55	57	2	6	55	56	1	10	46	43	−3

lationship to Simon and Friendly is reversed, with the five Kildew students showing less change than those at the other colleges.

In general, however, these analyses of two and four year change for similar subgroups of students at different colleges yielded two major findings: both the amounts and directions of change varied from college to college; the relative positions of the colleges remained roughly consistent for both two and four year change on most scales. Similar students who enter different kinds of colleges, do change differently. Institutional differences do make a difference to student development.*

Are different patterns of student change systematically related to differences among the colleges?

Relationships between two year and four year change, and college characteristics, were examined by (a) ranking the two groups of six colleges on magnitude of change, from plus to minus, (b) ranking the colleges on varied data concerning institutional characteristics, and (c) computing rank order correlation coefficients. For example, using the data on Table IX, ranks for Two Year Change on Autonomy are: Kildew–1, Classic–2, Elder–3, Simon–4, Friendly–5, and Stonewall–6. When these same institutions are ranked on the Practicality scores of the "College and University Environment Scales" their ranks are: Kildew–6, Classic–5, Elder–4, Friendly–3, Simon–2, and Stonewall–1. When these two sets of ranks, which are in almost perfect inverse relationship, are correlated, the resulting coefficient is –0.90, reported on Table X opposite the Practicality scale under Autonomy. This procedure was followed for each of the different scales for both two year and four year change, in relation to selected measures of institutional characteristics.

Findings are reported below for the four scales where patterns of change were most consistent: Autonomy, Impulse Expression, Complexity, and Practical Outlook; and for clusters of institutional characteristics which showed greatest internal consistency. Most of the data concerning institutional characteristics come from the "Experience of College Questionnaire" completed by samples of 80 to 180 students at each college selected from all four grade levels to reflect distributions of men and women in each grade.

It is important to note that the OPI scales are correlated with one another. Throughout these findings, for example, variables correlating positively with Autonomy, Impulse Expression, and Complexity almost always correlate negatively with Practical Outlook because Practical Outlook correlates negatively with the other three OPI scales. Institutional characteristics also tend to be inter-

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*The small N's for each subgroup increase the possibility of chance effects. But if chance was heavily operative, the internal consistency across different groups at two different points in time would be unlikely to occur.

TABLE X. Two and Four Year Change Correlated with College Emphasis and Student Characteristics

	Autonomy		Impulse Expression		Complexity		Practical Outlook	
	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr
College and University Environment Scales:								
Practicality	−0.90	−0.82	−0.57	−0.82	−0.67	−0.78	−0.99	−0.66
Community	0.14	−0.34	−0.60	−0.44	0.32	−0.14	−0.14	0.89
Awareness	0.83	0.55	0.30	0.43	0.54	0.42	−0.83	0.16
Propriety	−0.54	−0.51	−0.96	−0.57	−0.14	−0.78	0.54	0.60
Scholarship	0.77	0.25	−0.10	0.08	0.09	0.12	−0.78	0.52
Average OPI Scores for Entering Class:								
Autonomy	0.99	0.83	0.66	0.79	0.77	0.99	−0.96	−0.38
Impulse Expression	0.60	0.79	0.99	0.89	0.32	0.72	−0.60	−0.78
Practical Outlook	−0.94	−0.61	−0.40	−0.61	−0.66	−0.82	0.94	0.25
Proportion of Students Holding Primary Orientation at Entrance:								
Vocational	−0.88	−0.37	−0.51	−0.28	−0.43	−0.64	0.89	−0.04
Academic	0.24	−0.81	0.30	−0.87	−0.38	−0.58	−0.21	0.55
Collegiate	−0.43	−0.57	−0.88	−0.74	−0.20	−0.42	0.43	0.95
Nonconformist	0.77	0.83	0.93	0.78	0.49	0.99	−0.78	−0.38

Note: rho 0.77 = $p < 0.10$, 0.83 = $p < 0.04$, 0.94 = $p < 0.02$. Two and four year change each studied for two groups of six colleges. Four colleges were common to both groups; though membership overlaps, student groups within colleges differ from scale to scale.

correlated, so the sets of figures for different scales and different variables are not independent.

The "College and University Environment Scales" assess general college emphases, "climate," or "press." Two of these–Practicality, "a practical instrumental emphasis," and Propriety, "group standards of decorum are important"–showed consistent negative relationships with Autonomy, Impulse Expression, and Complexity, and positive relationships with Practical Outlook. The other scales showed no strong or consistent pattern.

The average scores for the total entering class and the orientations to college at entrance, both reflect general student characteristics pertinent to the areas of change examined. Scale score levels were related to change in the directions expected. For example, when the mean scores of the entering class were high on Autonomy, more change toward Autonomy occurred for students in the subgroups; when mean scores on Practical Outlook were high, there were smaller changes in Autonomy and greater changes in Practical Outlook. Among the four orientations to college, the proportion of students holding the Nonconformist Orientation bore the strongest relationships to change; Autonomy, Impulse Expression, and Complexity increased more, and Practical Outlook decreased, where the proportions of Nonconformists were relatively high. The Vocational and Collegiate Orientations consistently bore a relation to change opposite from the Nonconformists though the coefficients were lower. Mixed results were found for the proportions of students holding the Academic Orientation.

These data stuggest that when a college has a practical and instrumental emphasis combined with a mannerly, proper atmosphere, and enrolls relatively high proportions of students with a Vocational or Collegiate Orientation who score high on Practical Outlook, greater changes in Practical Outlook will occur and smaller changes in Autonomy, Impulse Expression, and Complexity, than at other colleges where these characteristics are not so prominent. When a college enrolls high proportions of Nonconformist students who score high on Autonomy and Impulse Expression, and low on Practical Outlook, Autonomy, Impulse Expression, and Complexity will increase more and Practical Outlook less.

Teaching practices and the study activities they fostered bore strong and consistent relationships to changes among students in the varied subgroups. Where heavy use of lectures occurred there was less change toward increased Autonomy, Impulse Expression, and Complexity, and greater change toward Practical Outlook. Where open arguments among students and between students and instructors more frequently occurred, and where students more often participated in decisions about course content and procedures, Autonomy, Impulse Expression, and Complexity increased more and Practical Outlook decreased.

At colleges where teachers usually lectured in class, students preparing for classes usually invested substantially more time Memorizing than in other more complex mental activities, so Memorizing also is negatively associated with

Autonomy, Impulse Expression, and Complexity, and positively associated with Practical Outlook. The amount of time invested in more complex study activities—interpreting, reorganizing, putting things in different terms and drawing inferences, applying concepts to new problems or situations, analyzing material to detect relationships among its parts, synthesizing ideas or information from diverse sources, evaluating evidence, theories, arguments, methods—was positively associated with increasing Autonomy, Impulse Expression, and Complexity, and negatively associated with Practical Outlook.

Where "intrinsic" reasons for study predominated—when study was pursued out of interest or enjoyment, to examine questions of concern, and to broaden

TABLE XI. Two and Four Year Change Correlated With Teaching, Study, and Reasons for Study

	Autonomy		Impulse Expression		Complexity		Practical Outlook	
Teaching Practices:	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr
Listening & taking notes	−0.60	−0.82	−0.77	−0.71	−0.03	−0.64	0.60	0.82
Making statements to the class	0.49	0.60	0.92	0.66	0.37	0.42	0.48	−0.95
Thinking about ideas presented	0.49	0.38	0.92	0.55	0.37	0.53	−0.48	−0.75
Lectures follow text	−0.94	−0.78	−0.61	−0.87	−0.77	−0.82	0.94	0.60
Instructor outlines lecture	−0.83	−0.31	−0.78	−0.37	−0.71	−0.72	0.83	0.33
Students argue openly with instructor	0.60	0.62	0.99	0.68	0.32	0.90	0.60	0.55
Students argue openly with students	0.72	0.73	0.93	0.78	0.32	0.93	−0.71	−0.61
Students Participate in Decisions about Course Content and Procedures	0.83	0.95	0.87	0.89	0.32	0.82	−0.83	−0.76
Mental Activities Studying for Class:								
Memorizing	−1.00	−1.00	−0.54	−0.94	−0.60	−0.78	0.00	0.68
Interpreting	0.49	0.60	0.92	0.78	0.37	0.62	−0.48	−0.85
Applying	0.49	0.49	0.92	0.72	0.37	0.42	−0.48	−0.81
Analyzing	0.43	0.72	0.93	0.83	0.20	0.58	−0.43	−0.95
Synthesizing	0.60	0.83	0.99	0.95	0.32	0.82	−0.60	−0.64
Evaluating	0.26	0.38	0.84	0.55	0.26	0.55	−0.26	−0.77
Reasons for Studying:								
Interest and enjoyment	0.00	0.48	0.59	0.42	0.60	0.88	−1.00	−0.31
Questions of concern	0.94	0.66	0.70	0.60	0.37	0.96	−0.94	−0.24
Broaden general knowledge	0.53	0.89	0.53	0.78	0.24	0.93	−0.47	−0.61
Vocationally useful	0.73	−0.60	0.66	−0.54	0.07	−0.92	−0.70	0.30
Get a good grade	−1.00	−0.95	−0.54	−0.87	−0.60	−0.82	0.00	0.60
Finish a requirement	−0.83	−0.90	−0.16	−0.75	−0.16	−0.81	0.83	0.45

Note: rho 0.77 = $p < 0.10$, 183 = $p < 0.04$, 0.94 = $p < 0.02$. Two and four year change each studied for two groups of six colleges. Four colleges were common to both groups; though membership overlaps, student groups within colleges differ from scale to scale.

the pool of general knowledge—there were greater increases in Autonomy, Impulse Expression and Complexity, and smaller increases in Practical Outlook. Where "extrinsic" reasons were more often of primary importance—to get a good grade or to finish a requirement for graduation—opposite relationships occurred.

These findings suggest that, when lectures, memorizing, and extrinsic motivations for study predominate, Practical Outlook increases more and Autonomy, Impulse Expression, and Complexity increase less. When greater proportions of class time are given to open exchange among students and between students and instructors, when class assignments more often foster the exercise of more complex mental activities, and study more often rests on intrinsic motives, Autonomy, Impulse Expression, and Complexity tend to increase and Practical Outlook to decrease.

The amount of time students invest in varied activities varies from college to college. Some uses of time are associated with changes and others are not. Where more hours are spent reading for pleasure and talking informally with others, there are greater increases in Autonomy, Impulse Expression, and Complexity and smalller increases in Practical Outlook. Time spent watchingTV bears an opposite relation to change. Hours spent studying for class or playing games are not associated with change.

Student-faculty relationships reflected strong and consistent correlations with change. At colleges where contacts with faculty members outside of class occurred more frequently, Autonomy, Impulse Expression, and Complexity increased and Practical Outlook decreased. It's worth noting that the amount of time spent in conversations with advisors or with members of the faculty and administration in general seems to be less important than simple frequency and diversity of contact. The most developmentally fruitful arrangements, therefore, provide students with ready access to diverse faculty members, even if only for brief contacts. Then problems can be considered when they arise, interest can be encouraged as they emerge, pleasures and excitements can be shared while they are fresh.

It seems clear that the different patterns of change which appear when subgroups of similar students are examined in different institutions, are systematically related to a variety of college characteristics. Though it is not clear from these analyses which relationships identify characteristics influential in their own right, and which relationships simply identify characteristics which do not have direct influence but are simply associated with those that do. The internal consistency among clusters of findings, however, does suggest that personality development in college is influenced not only by the general emphasis or climate of the college and by the characteristics of the other students, but that personality development also is influenced by the concrete experiences and behaviors generated by varied teaching practices and study activities and by the

TABLE XII. Two and Four Year Change Correlated with Number of Hours Per Week Spent in Varied Activities

	Autonomy		Impulse Expression		Complexity		Practical Outlook	
	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr
Studying for class	0.37	0.43	−0.40	0.55	0.37	0.56	−0.37	0.02
Reading for pleasure	0.69	0.89	0.99	0.95	0.32	0.77	−0.60	−0.87
Talking informally with others	0.66	0.73	0.93	0.79	0.14	0.93	−0.66	−0.61
Watching TV	−0.92	−0.37	−0.15	−0.24	−0.60	−0.64	0.83	−0.04
Playing chess, cards, checkers, etc.	−0.43	0.02	0.42	0.13	−0.37	−0.20	0.43	−0.21

Note: rho 0.77 = $p < 0.10$, 0.83 = $p < 0.04$, 0.94 = $p < 0.02$. Two and four year change each studied for two groups of six colleges. Four colleges were common to both groups; though membership overlaps, student groups within colleges differ from scale to scale.

diversity and frequency of contact between students and faculty (Chickering, 1970, 1971a, 1971b, 1972).

Summary and General Implications

Studies of net change found that:

1. Changes occur in most of the areas measured by the OPI.
2. The directions and amounts of net change are similar for both men and women, and similar for the diverse groups of entering freshmen.
3. The directions and amounts of net change also were similar at all the Pro-

TABLE XIII. Two and Four Year Change Correlated with Student-Faculty Relationships

	Autonomy		Impulse Expression		Complexity		Practical Outlook	
	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr	2 Yr	4 Yr
Student-Faculty Contact:								
Number of faculty seen outside of class	0.83	0.83	0.87	0.78	0.32	0.99	−0.83	−0.38
Number of conversations outside of class	0.77	0.72	0.93	0.78	0.49	0.93	−0.77	−0.56
Amount of Time in Conversations with Faculty and Administration in General about:								
Formal academic arrangements	−0.37	0.03	0.33	0.15	−0.37	0.30	0.37	0.05
Future educational and vocational plans	0.04	0.43	0.58	0.38	−0.20	0.82	−0.01	−0.07
Personal problems	−0.03	0.55	0.44	0.49	0.03	0.90	−0.26	−0.44
Topics in academic field of faculty member	0.60	0.72	0.99	0.78	0.32	0.93	−0.60	−0.61
Campus events	0.54	0.78	0.81	0.60	−0.14	0.79	−0.54	−0.61
General conversations	0.47	0.49	0.89	0.55	0.33	0.85	−0.44	−0.51
Amount of Time in Conversations with Advisor:								
Formal academic arrangements	0.14	0.60	0.56	0.66	0.37	0.85	−0.14	−0.24
Future educational and vocational plans	0.37	0.32	0.10	0.38	0.20	0.76	−0.37	−0.30
Personal problems	0.37	0.49	0.10	0.55	0.20	0.85	−0.37	−0.51
Topics in academic field of advisor	0.60	0.49	0.99	0.55	0.32	0.85	−0.60	−0.51
Campus events	0.49	0.38	0.76	0.26	0.09	0.50	−0.48	−0.61
General conversations	0.60	0.43	0.81	0.32	0.03	0.76	−0.60	−0.40

Note: rho 0.77 = $p < 0.10$, 0.83 = $p < 0.04$, 0.94 = $p < 0.02$. Two and four year change each studied for two groups of six colleges. Four colleges were common to both groups; though membership overlaps, student groups within colleges differ from scale to scale.

ject colleges, despite the dramatic differences among them.

4. The scales showing largest and most consistent changes were Autonomy, Impulse Expression, Personal Inegration, Estheticism, Complexity, Religious Orientation (Liberalism), and Thinking Introversion, where scores rose, and Practical Outlook, where scores dropped. Smaller and less consistent changes occurred on Altruism, Theoretical Orientation, Anxiety Level, and Response Bias, where scores rose, and on Masculinity-Femininity, where scores dropped. No consistent changes occured on Social Extroversion.

5. Changes in particular attitudes, beliefs, and behaviors also were common to many of the different types of students at the different colleges.

Studies of change among subgroups of similar students at different colleges found that:

1. Similar students attending different colleges reflected different patterns of change.

2. At some colleges Autonomy, Impulse Expression, and Complexity consistently showed substantial increases while Practical Outlook decreased; at other colleges smaller increases in Autonomy, Impulse Expression, and Complexity were accompanied by smaller decreases in Practical Outlook.

3. Changes among the subgroups were systematically related to institutional differences in general college climate or emphasis, student characteristics, teaching practices and study activities, time spent reading for pleasure, talking informally with others, and watching TV and student-faculty relationships.

The detailed findings for individual colleges which underlie these generalizations have implications for each of those institutions. Examination of those details and consideration of their particular implications is beyond the scope of this report and is left to the individual colleges. However, there are some general implications worth attention because they suggest fundamental shifts in how we think about college influences on personality development.

It is clear that the college student is no "tabula rasa"; he is no clay for the potter, no vessel to be filled, no lamp to be lighted. He's already lit. Usually the burners are fired up and he's maintaining orbits around several centers of gravity. When he moves into college as a freshman he brings with him—along with his Webster's Collegiate Dictionary and Tensor Lamp—his mother, father, and two older friends of the family, a girl friend and a set of high school buddies. He brings strengths and weaknesses, prides and prejudices, clarities and confusions, and a lot of unfinished business. The business to be finished typically includes not only the development of intellectual and interpersonal competence, but also developing autonomy and learning better ways to manage sexual and agressive impulses, becoming freer with diverse kinds of persons, clarifying identity, and developing sharper purposes and greater integrity (Chickering, 1969).

Most students move toward institutions whose purposes and programs fit their own interests and inclinations and whose students and faculty hold values and attitudes similar to their own. This process of self-selection and institutional selection creates a comfortable fit between the college and the person for substantial numbers of students. Though the college may not act "in loco parentis," it does act "in loco uterus," providing a warm and supportive setting which insulates most students from unduly disruptive outside influences.

Under these conditions of comfortable fit, personality development proceeds along the vectors of change set by the general cultural and genetic forces operating in our society; most students in most colleges work on the major areas of unfinished business mentioned above. Across the country there are a variety of student-college types, and in each type these common developmental tasks are pursued in somewhat different fashion.* But because the institutional differences are accompanied by corresponding differences among the students, the patterns of general development are about the same among the different types.

One major model for "college influence," therefore, is the womb. The diverse colleges provide safe havens and appropriate nourishment for the diverse students who characterize a pluralistic American society. College attenders become more autonomous, more flexible, more complex, and less materialistic, more aware of their own emotions and able to express them in thought and action, more tolerant of ambiguity, less dogmatic, more intellectually curious. Students who do not attend college, change less in these areas or move in contrary direction. (Plant et al., 1965). So wombs are good things. Without them, most of us would not be here. And without the protection and nourishment many colleges offer, most seniors would not have become what they are at graduation.

But there is another kind of "college influence,"—and in this case, where it was not before, the term "impact" is appropriate. At every institution there are two kinds of Misfits: those whose development has not yet proceeded to the general level at which the college operates, and those whose development has gone beyond the operating level of the college. Misfitting can occur along many dimensions. A student's intellectual competence and breadth of information may be so limited that he finds it difficult to cope with the academic program, or his competence and information level may be so high that course demands, classes, and other students contribute little. Students may be more liberal than most others at the college, or more conservative, more cultually sophisticated or less, more autonomous or more dependent. The proportions of Misfits in any college, and their distinguishing characteristics, will vary depending upon how selective the college is, and upon its criteria for selection.

By examining subgroups of similar students in different colleges, we were, in

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* Although the results reported above are for small colleges only, they are consistent with those which have been found for many different kinds of colleges and universities.

effect, studying persons who were Misfits in some colleges but not at others, and persons who deviated from the norm in different ways at different colleges. The special impact on such persons of institutional differences in educational practices, in student-faculty relationships, in general atmosphere, and in student characteristics was quite clear.

The members of these subgroups were Misfits who persisted through the four years and changed as a consequence. Others like them, however, did not persist. They left, to transfer to another college, to go to work, or simply to roam through varied settings. It is not pure coincidence that the characteristics which most sharply distinguish leavers from stayers are just those characteristics on which greatest change occurs. Compating Project college leavers and stayers reveals the same differences found consistently in other studies. Compared with those who stay, leavers are more autonomous, more impulsive, more complex, more tolerant of ambiguity, more creative (Hannah, 1969a).

But the important thing is that "higher" or "lower" in such areas depends entirely upon the college and upon the student's position in relation to it. For example, at WJB the leavers scored significantly higher than the stayers on such measures, but the leavers' scores were still lower than 70% of the 7000 freshmen in the norms group. So compared with other college students across the country these leavers are relatively cautious, conservative, conforming, and dependent. But compared with their peers who stayed they are outstanding in their impulsiveness, complexity, and autonomy. At the other end of the spectrum, the scores of Kildew girls who stayed were higher than 85% of the norms group. These girls, by any standard are highly creative, impulsive, complex, and autonomous. Yet the girls who left scored significantly higher. The other Project colleges fall between these two extremes but the pattern remains the same. At each institution it is the fit between the student and his institution which makes a difference to his leaving or staying, not his general level in relation to the total population (Hannah, 1969b).

Because the characteristics which distinguish leavers from stayers are the areas in which greatest changes occur, we must recognize that, under current institutional conditions, continued growth for these Misfits may require their premature departure to different settings. And we must recognize that such moves toward more challenging and fruitful environments often are healthy steps. Taking charge of one's own existence in this way is to be valued and fostered, not decried and curtailed.

The other major model of college influence or impact, therefore, is the Misfit. When a college helps Misfits survive, and through those experiences develop in ways valued both by them and by the institution, it makes a substantial contribution. It also makes a contribution when it helps those Misfits who have to move elsewhere do so with clearer purposes and greater confidence. If however, the

college confirms in such leavers feelings of insecurity, instability, or idiosyncracy, if it will not recognize their special situation and respond to it, then there is great potential for damage.

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